PIPELINE AUDIT AND HEALTH REPORT

Untitled pipeline

Generated 2026-10-13 19:48 | simulated 2.0 h | seed 42 | 10 replications

HEALTH SCORE

70 /100

Grade C

Acceptable: workable, but there are clear risks to address before scaling up.

Critical 0 | High 1 | Medium 3 | Low 1 | Info 1

The score starts at 100 and loses points for each finding. The deductions are listed in Appendix B.

THROUGHPUT	COMPLETED	ARRIVALS LOST
202.5 units/h	405 units	31.8%

AVERAGE WIP	CYCLE TIME	BOTTLENECK
5.1 units	91 s	milling_1

Executive summary

Primary bottleneck: 'milling_1' (56% busy, 19% down, index 79). 31.8% of arrivals are lost. Run what-if to measure which lever recovers the most throughput.

What to fix first

- High (arrivals): Add capacity at the constraint or enlarge the buffer in front of it. Measured best lever: 10% faster processing at 'milling_1' (+2.6% throughput).
- Medium (milling_1): Shorten repairs (spares, on-call technicians) or introduce preventive maintenance. A 20% shorter repair time is a what-if lever.
- Medium (milling_1, assembly_1): Add a parallel machine or a buffer of finished work in front of the downstream stations so they can ride through a repair.
- Low (throughput): Use a longer run if you need tighter estimates.

Best measured improvement

Best lever: 10% faster processing at 'milling_1' -> +6.6 units/hour (2.6%), measured over 3 seeds with common random numbers.

Findings

Each finding names where the problem is, why it matters, and what to do. Severity reflects the impact on throughput, correctness or confidence in the results.

SEVERITY	AREA	WHERE	ISSUE AND RECOMMENDED ACTION	POINTS
High	Flow	**arrivals**	31.8% of arrivals are lost because a buffer is full. Action: Add capacity at the constraint or enlarge the buffer in front of it. Measured best lever: 10% faster processing at 'milling_1' (+2.6% throughput).	-15

SEVERITY	AREA	WHERE	ISSUE AND RECOMMENDED ACTION	POINTS
Medium	Reliability	**milling_1**	Down 19% of the time (3 failures). Action: Shorten repairs (spares, on-call technicians) or introduce preventive maintenance. A 20% shorter repair time is a what-if lever.	-3
Medium	Reliability	**assembly_1**	Down 14% of the time (2 failures). Action: Shorten repairs (spares, on-call technicians) or introduce preventive maintenance. A 20% shorter repair time is a what-if lever.	-3
Medium	Reliability	**milling_1, assembly_1**	Every unit passes through these stations, and they can fail. A single failure stops the whole plant. Action: Add a parallel machine or a buffer of finished work in front of the downstream stations so they can ride through a repair.	-8
Low	Confidence	**throughput**	Throughput 95% CI is +-9.0% of the mean. Action: Use a longer run if you need tighter estimates.	-1
Info	Capacity	**milling_1**	Constraint of the line: 75% utilised. Action: Improvements elsewhere will not raise throughput until this station is relieved. Measured best lever: 10% faster processing at 'milling_1' (+2.6% throughput).	

Performance by node

Share of the measured window each node spent busy, blocked (finished part cannot leave), down (failed) and starved (waiting for work).

Legend: green = busy, amber = blocked, red = down, grey = starved or idle.

NODE	KIND	STATE MIX	BUSY %	BLOCKED %	DOWN %	AVG / MAX WIP
ingestion	source	-	-	-	-	-
milling_1 bottleneck	machine		55.9	10.8	19.2	3.3 / 6
assembly_1	machine		50.5	0.0	13.9	1.4 / 5
inspection_1	machine		30.8	0.0	0.0	0.4 / 4
__shipped	sink	-	-	-	-	-

Work in process over time

Reliability and robustness

The pipeline was simulated 10 times with different random seeds. A wide interval means the result depends strongly on luck.

METRIC	MEAN	95% CONFIDENCE	STD DEV	RELATIVE
Throughput	248.75 units/h	+- 22.34	31.23	9.0%
Average WIP	4.05 units	+- 0.45	0.64	11.2%
Cycle time	60.25 s	+- 12.60	17.61	20.9%
Arrivals lost	15.20 %	+- 7.00	9.79	46.0%

No replication deadlocked.

Failure model

NODE	MTBF	MTTR	AVAILABILITY	OBSERVED DOWN	FAILURES
milling_1	60 min	5.0 min	92.3%	19.2%	3
assembly_1	40 min	3.3 min	92.3%	13.9%	2

Single points of failure

All units must pass through: milling_1, assembly_1, inspection_1. If one of these stops, output stops.

Model validity checks

CHECK	RESULT	DETAIL
Unit conservation	PASS	created 597 = completed 405 + scrapped 0 + lost 190 + in system 2
Little's Law	PASS	L = 5.13, lambda x W = 5.12 (gap 0.2%)
Run length	PASS	window = 79 x cycle time (30x or more recommended)
Warm-up	WARN	no warm-up: start-up transients are included in averages
Deadlock-free	PASS	reference run completed

What-if analysis

Each lever was applied to one of the most constrained stations and the whole pipeline re-simulated over at least 8 hours with 3 identical random seeds per scenario. Baseline throughput: 254.2 units/h.

NODE	CHANGE	UNITS/H	CHANGE	CHANGE %
milling_1	10% faster processing	260.8	**+6.6**	+2.6%
milling_1	+2 buffer slots	259.0	**+4.8**	+1.9%
milling_1	20% shorter repairs	258.0	**+3.8**	+1.5%
assembly_1	20% shorter repairs	258.0	**+3.8**	+1.5%
assembly_1	+2 buffer slots	255.1	**+0.9**	+0.3%

NODE	CHANGE	UNITS/H	CHANGE	CHANGE %
assembly_1	10% faster processing	255.0	**+0.8**	+0.3%
inspection_1	+2 buffer slots	254.2	**+0.0**	+0.0%
inspection_1	10% faster processing	254.2	**+0.0**	+0.0%

Best lever: 10% faster processing at 'milling_1' -> +6.6 units/hour (2.6%), measured over 3 seeds with common random numbers.

Appendix A: node inventory

Effective parameters after node-type templates are applied. Parameters not listed use the node type's defaults.

NODE	TYPE	EFFECTIVE PARAMETERS
ingestion	source	interarrival=exponential(12.00)
milling_1	machine	capacity=5, mtbf=3600.00, mttr=300.00, process=normal(10.00,1.50)
assembly_1	machine	capacity=4, mtbf=2400.00, mttr=200.00, process=normal(9.00,1.00)
inspection_1	machine	capacity=3, process=exponential(5.00)
__shipped	sink	(defaults)

Appendix B: how the score is computed

Health score = 100 minus the sum of the penalties below, never less than 0. Grades: A 90 and above, B 80 to 89, C 70 to 79, D 60 to 69, F below 60. Penalties are capped per category so that one repeated issue cannot dominate the score. Informational findings carry no penalty.

SEVERITY	WHERE	FINDING	POINTS
High	**arrivals**	31.8% of arrivals are lost because a buffer is full.	-15
Medium	**milling_1**	Down 19% of the time (3 failures).	-3
Medium	**assembly_1**	Down 14% of the time (2 failures).	-3
Medium	**milling_1, assembly_1**	Every unit passes through these stations, and they can fail. A single failure stops the whole plant.	-8
Low	**throughput**	Throughput 95% CI is +-9.0% of the mean.	-1
	Total deductions		**-30**
	Health score		**70**

Method: the pipeline is simulated as a discrete-event system; the bottleneck index is utilisation (busy plus down time) with small weights for starved downstream and blocked upstream neighbours. What-if results are measured, not estimated. Confidence intervals use Student's t over independent replications.